

07 — Maximum power transfer

Class notes p.13–14, p.16–17, set 2 p.14 · MTE syllabus item 5 (DC and AC) · 2 board question sets + 1 textbook AC problem · needs 06

Map

1 Power in the load

→

2 The condition $R_L = R_{Th}$

→

3 P_{max} and the 50 % efficiency

→

4 AC loads: conjugate match

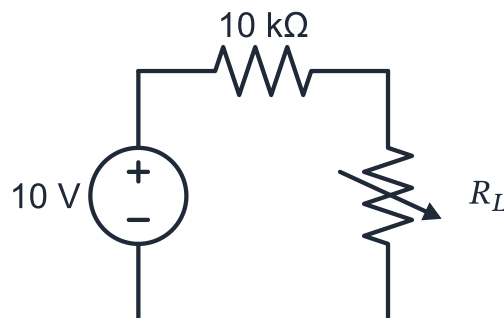
The questions this file answers

- **Class p.16–17** 10 V source, 10 k Ω , variable R_L : for what R_L is power maximum? Total power from the source? Power in R_{Th} ? Efficiency? Voltage across the load? How do the load's voltage and power compare with the source's?
- **Class p.17** 10 V source, 5 k Ω : find R for maximum power, the maximum power, and the efficiency.

Build

CLASS P.16

For what value of load resistance will maximum power be delivered? Then: determine the total power delivered by the source.



The board circuit: a 10 V source with 10 k Ω in series, driving a variable load.

GUESS FIRST

Make R_L tiny, then huge. What happens to the load's power at each extreme? So where must the maximum be?

1 · Power in the load

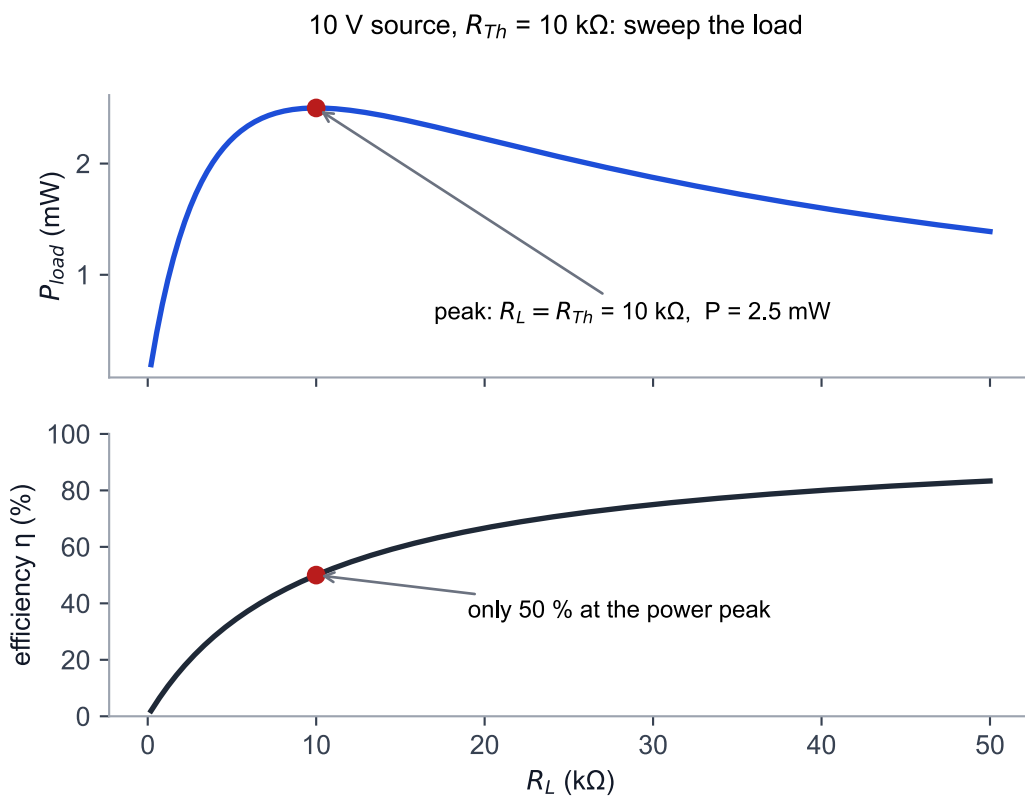
From 06, every linear source network is V_{Th} in series with R_{Th} , so this circuit is already in that form. The load current and power:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \quad P_L = I_L^2 R_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

CLASS P.13

The first attempt on the board used $P_L = V_{Th}^2 / (R_{Th} + R_L)$, which is wrong; it was crossed out and redone with $P_L = I_L^2 R_L$ (starred). Use the starred one.

The Guess answer: tiny R_L gives big current but almost no voltage, so $P \approx 0$. Huge R_L gives big voltage but almost no current, so $P \approx 0$. The power must peak somewhere in between:



Top: the load's power peaks exactly at $R_L = R_{Th}$. Bottom: efficiency keeps rising with R_L and is only 50 % at that peak.

2 · The condition (class p.13–14 derivation)

Set $\frac{dP_L}{dR_L} = 0$. Quotient rule:

$$\frac{dP_L}{dR_L} = V_{Th}^2 \cdot \frac{(R_{Th} + R_L)^2 - R_L \cdot 2(R_{Th} + R_L)}{(R_{Th} + R_L)^4} = 0$$

The numerator must be zero:

$$(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0 \Rightarrow R_{Th}^2 + 2R_{Th}R_L + R_L^2 - 2R_{Th}R_L - 2R_L^2 = 0 \Rightarrow R_{Th}^2 = R_L^2$$

$$R_L = R_{Th}$$

(The negative root is not a resistance. The curve above shows it is a maximum, not a minimum.)

Answer, part 1: $R_L = R_{Th} = 10 \text{ k}\Omega$.

3 · P_{max} , source power, efficiency

Put $R_L = R_{Th}$ into P_L :

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

For the board circuit, at $R_L = 10 \text{ k}\Omega$:

Quantity	Working	Value
Current	10 V / 20 k Ω	0.5 mA
Total power from the source	$V \cdot I = 10 \times 0.5 \text{ mA}$	5 mW
Power in the load	$I^2 R_L = (0.5 \text{ mA})^2 \times 10 \text{ k}\Omega = V_{Th}^2 / 4R_{Th}$	2.5 mW
Power in R_{Th}	same current, same resistance	2.5 mW
Efficiency η	$P_{load} / P_{total} = 2.5 / 5$	50 %
Voltage across the load	$0.5 \text{ mA} \times 10 \text{ k}\Omega$	5 V = $V_s/2$
Load vs source		$V_L = V_s/2, P_L = P_s/2$

CLASS P.16

The notes read " $I \times V = 0.5 \text{ mW}$ " with no visible decimal point. The value is **5 mW**. The 50 % efficiency on p.17 is right either way.

Maximum power is not maximum efficiency. $\eta = \frac{R_L}{R_{Th} + R_L}$ keeps rising as R_L grows (bottom plot), while P_L falls. So:

- **communication / signal circuits** match $R_L = R_{Th}$: a weak signal needs every bit of power delivered, and 50 % loss is acceptable;
- **power systems** keep $R_L \gg R_{Th}$: wasting half the generated power would be unacceptable.

The 50 % refers to the Thévenin equivalent. The power actually lost inside the original, full network can differ.

CHECK 1

- (a) Why is exactly half the power lost at the match? Use the fact that R_{Th} and R_L carry the same current.
- (b) The class p.17 question: 10 V with 5 k Ω . Find R , P_{max} , η .

4 • AC loads: the conjugate match

PREDICT

A source has $Z_{Th} = 3 + j4 \Omega$. Its reactance $j4 \Omega$ doesn't absorb power, but it does limit the current. What reactance should the load add?

For impedances (class p.16; set 2 p.14: " $Z_L = Z_{Th}^*$, $*$ = complex conjugate, $i \rightarrow -i$):

$$Z_L = Z_{Th}^* \quad (\text{the complex conjugate: } j \rightarrow -j)$$

- $R_L = R_{Th}$: the resistance match, as before.
- $X_L = -X_{Th}$: the load's reactance **cancels** the source's, so the loop is purely resistive and the current is as large as possible.

For the Predict: $Z_L = 3 - j4 \Omega$. Then $P_{max} = \frac{|V_{Th}|^2}{4R_{Th}}$, with V_{Th} as an **rms** value (with a peak value it is $\frac{|V_{Th}|^2}{8R_{Th}}$).

TEXTBOOK

The AC network of 06 §6 has $V_{Th} = 7.07 \angle -45^\circ$ V (peak) and $Z_{Th} = 500 - j500 \Omega$. Find the load for maximum power, and that power.

- **Load:** $Z_L = Z_{Th}^* = 500 + j500 \Omega$ (500 Ω with an inductive reactance of 500 Ω ; at $\omega = 1000$ that is $L = 0.5$ H).
- **Loop:** $Z_{Th} + Z_L = 1000 \Omega$, purely resistive: the reactances cancel. $I = \frac{7.07}{1000} = 7.07$ mA peak = 5 mA rms.
- **Power:** $P = I_{rms}^2 R_L = (5 \text{ mA})^2 \times 500 \Omega = 12.5 \text{ mW}$. Same by formula: $V_{rms} = 5 \text{ V}$, $\frac{25}{4 \times 500} = 12.5 \text{ mW}$.

A computer sweep over every load $R + jX$ gives its peak at exactly $R = 500 \Omega$, $X = +500 \Omega$, 12.5 mW.

Exam form

Statement. A linear network delivers maximum power to a resistive load when $R_L = R_{Th}$. The maximum power is $P_{max} = V_{Th}^2 / (4R_{Th})$. For an AC network with a complex load, the condition is $Z_L = Z_{Th}^*$.

Derivation. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; set $dP_L / dR_L = 0 \Rightarrow R_L = R_{Th}$ (step 2).

At the match. $\eta = 50 \%$, $V_L = V_{Th} / 2$, $P_L = P_S / 2$.

Traps

- Using $P = V_{Th}^2 / (R_{Th} + R_L)$ (the crossed-out attempt on class p.13).
- Saying "maximum power means maximum efficiency".
- In the AC case, matching $Z_L = Z_{Th}$ instead of the **conjugate**.
- Forgetting to find R_{Th} first when the source network is more than one resistor: use 06.

Self-test

1. Derive $R_L = R_{Th}$ (three lines).
2. $V_{Th} = 12\text{ V}$, $R_{Th} = 3\text{ k}\Omega$. Find P_{max} and the source power at that point.
3. Why do power companies not operate at maximum power transfer?
4. $Z_{Th} = 50 - j20\ \Omega$. Load for maximum power? If $V_{Th} = 10\text{ V rms}$, what is P_{max} ?
5. In the 06 board circuit ($V_{Th} = 20/3\text{ V}$, $R_{Th} = 8/3\text{ k}\Omega$), what R_L gets maximum power, and how much?

Answers

Check 1. (a) At the match, $R_{Th} = R_L$ and both carry the same current, so $I^2 R_{Th} = I^2 R_L$: equal power in each, i.e. half of the total. (b) $R = 5 \text{ k}\Omega$; $P_{max} = 10^2 / (4 \times 5 \text{ k}\Omega) = 5 \text{ mW}$; $\eta = 50 \%$ (the source supplies $10 \text{ V} \times 1 \text{ mA} = 10 \text{ mW}$).

Self-test.

1. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; the numerator of the derivative is $(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0$; this gives $R_{Th}^2 = R_L^2$, so $R_L = R_{Th}$.
2. $P_{max} = 144 / 12 \text{ k} = 12 \text{ mW}$; $I = 12 / 6 \text{ k} = 2 \text{ mA}$; $P_s = 24 \text{ mW}$.
3. Efficiency would be only 50 %: half of the generated power lost in the source and lines.
4. $Z_L = 50 + j20 \Omega$; $P_{max} = 100 / (4 \times 50) = 0.5 \text{ W}$.
5. $R_L = 8/3 \text{ k}\Omega \approx 2.67 \text{ k}\Omega$; $P_{max} = (20/3)^2 / (4 \times 8/3) = 4.17 \text{ mW}$.